

Math 1 • Spec remediation review

Only replaced / fixed / rewritten questions from Mocks B–E (Mock A unchanged). For checking — not a full paper.

B7 • REPLACE

log_2 out of spec (MM5)

The graphs of the functions $y = (k - 1)x^2 + 6x$ and $y = -k - 7$ intersect at exactly one point. What is the sum of all possible real values of the constant k ?

- A -6
- B -5
- C 1
- D 2
- E 3

B18 • REPLACE

arithmetic-series sum (MM2.2)

How many integer values of x are there such that the value of the expression

$$\frac{3x + 31}{3x + 1}$$

is an integer?

- A 2
- B 4
- C 8
- D 16

B19 • REPLACE

flagged for replacement

A fair six-sided die is rolled three times. The result of each roll, d , is used to form a quadratic equation $x^2 + dx + 9 = 0$.

What is the probability that at least one of the three equations has real roots?

- A** $\frac{1}{216}$
- B** $\frac{91}{216}$
- C** $\frac{125}{216}$
- D** $\frac{1}{2}$
- E** $\frac{26}{27}$
- F** $\frac{215}{216}$

C3 • REPLACE

log_2 out of spec (MM5)

A square has side length x cm. A rectangle is formed by doubling one dimension of the square and decreasing the other dimension by 4 cm. The area of the resulting rectangle is 15 cm^2 less than the area of the original square.

What is the perimeter of the square?

- A** 12 cm
- B** 14 cm
- C** 15 cm
- D** 16 cm
- E** 20 cm

C11 • FIX

supply sphere/cone volume formulae

A solid metal sphere of diameter 6 cm and a solid metal cylinder of diameter 8 cm and height 3 cm are melted down. The molten metal is used to cast a number of identical solid cones, each with diameter 2 cm and height 3 cm.

You may use the formulae

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3, \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h.$$

Assuming no metal is lost, how many complete cones can be made?

- A** 28
- B** 36
- C** 48
- D** 75
- E** 84
- F** 120

C21 • REPLACE

difficulty/style (tangent circles)

A tank is filled by a tap and emptied by a drain. The drain takes 10 minutes longer to empty the full tank than the tap takes to fill the empty tank.

Starting with an empty tank, both the tap and the drain are opened. After 45 minutes, the tank is $\frac{3}{4}$ full. How many minutes does it take for the tap to fill the tank when the drain is closed?

- A** 20
- B** 25
- C** 30
- D** 55
- E** 60

D13 • REPLACE*heavy combinatorics*

Three fair, six-sided dice are rolled. Let p be the probability that the sum of the numbers obtained is 6, and let q be the probability that the sum of the numbers obtained is 7.

What is the value of $\frac{q}{p}$?

- A 1
- B $\frac{7}{6}$
- C $\frac{4}{3}$
- D $\frac{3}{2}$
- E $\frac{5}{3}$

D16 • FIX*supply cone volume formula*

Two cones have the same volume. The height of the second cone is twice the height of the first cone.

You may use the formula

$$V = \frac{1}{3}\pi r^2 h.$$

If the base radius of the first cone is r , what is the base radius of the second cone?

- A $\frac{r}{\sqrt{2}}$
- B $\frac{r}{2}$
- C $\frac{r}{4}$
- D $r\sqrt{2}$
- E $2r$
- F $4r$

D22 • REPLACE

general circle equation (MM3.2)

A composite solid cylinder is formed from two metals, P and Q. The volume of metal P is 60 cm^3 and the volume of metal Q is 90 cm^3 . The density of metal P is 5 g/cm^3 and the density of metal Q is 2 g/cm^3 .

The cylinder has a uniform internal cross-sectional area of 10 cm^2 . The metals form two distinct layers, with the denser metal at the bottom.

What is the total mass of the horizontal slice of the cylinder between 4 cm and 8 cm from the base?

- A** 14 g
- B** 80 g
- C** 100 g
- D** 128 g
- E** 140 g
- F** 480 g

E6 • REWRITE

clarify single-piece cutting

A solid cuboid has side lengths 2^a , 2^b , and 2^c , where a , b , and c are integers such that $a > b > c > 0$.

The following process is applied repeatedly to a single piece.

- Cut that piece in half by a plane perpendicular to its longest edge. If two or more edges share the longest length, choose exactly one of them.
- Keep exactly one of the two resulting pieces and discard the other.
- Continue with the kept piece.

The process stops as soon as the kept piece is a cube.

Let N be the total number of cuts required, and let S be the side length of the final cube.

Which pair of expressions for N and S is correct?

- A** $N = a - b, \quad S = 2^b$
- B** $N = a - c, \quad S = 2^c$
- C** $N = a + b - c, \quad S = 2^c$
- D** $N = a + b - 2c, \quad S = 2^c$
- E** $N = a + b - 2c, \quad S = 2^{3c}$

E25 • REPLACE*radians / sector (MM4.2)*

Let the function f be defined by $f(x) = x^2 - 2$.

The function $f^n(x)$ is defined by $f^1(x) = f(x)$ and $f^{n+1}(x) = f(f^n(x))$ for $n \geq 1$.

Find the number of distinct real solutions to the equation $f^5(x) = 0$.

- A** 5
- B** 10
- C** 16
- D** 25
- E** 32
- F** 64

E26 • REPLACE*modulus inequality (Math 2)*

The quantities x , y , P , Q and R in a physical system are related by the equation:

$$y = \frac{Px + Q}{x - R}$$

Which expression gives x in terms of y , P , Q and R ?

- A** $x = \frac{yR + Q}{P - y}$
- B** $x = \frac{Q + R}{y - P}$
- C** $x = \frac{yR - Q}{y - P}$
- D** $x = \frac{yR + Q}{y - P}$
- E** $x = \frac{yR + Q}{y + P}$